Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 11 Laplace Transform Applications

discuss in class

1. (a) Solve $y(t)$ for the first order ODE using the method of Laplace Transform

$$y' - 2y = u(t), \qquad y(0) = 1.$$

Is the response stable? What caused it to be stable/unstable?

(b) Solve $y(t)$ for the linear second order ordinary differential equation

$$y'' - 2y' - 3y = u(t), \qquad y(0) = 0, \qquad y'(0) = 0.$$

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2. Find the time function of $i(t)$ after the switch S is closed at $t = 0$. The capacitor is fully discharged before the switch is closed.

If $i(t)$ is considered as the output, and $v_{in}(t)$ is considered as the input. Find the Transfer Function of this linear system.

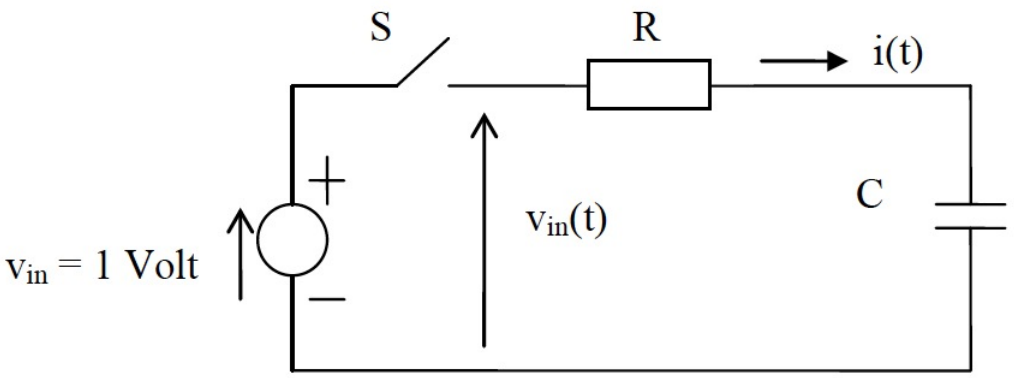


Figure 1

3. (a) Write the formula for $f(t) * g(t)$, where $*$ is the convolution operator.

(b) [Convolution Theorem]
What is $L\{f(t) * g(t)\}$ equal to?
What is $L^{-1}\{F(s)G(s)\}$ in the time domain?

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(c) The output $y(t)$ of a linear system, with impulse response of $h(t)$, subjected to an input function $x(t)$ is given by $y(t) = h(t) * x(t)$.

If $x(t) = u(t)$ and $H(s) = \dfrac{4}{s^2 + 2s + 5}$,

(i) find $h(t)$, the impulse response of $H(s)$

(ii) solve for $y(t)$ by using the Transfer Function approach.

Answers

1. (a) $-0.5u(t) + 1.5e^{2t}u(t)$ (b) $\left(-\frac{1}{3} + \frac{1}{4}e^{-t} + \frac{1}{12}e^{3t}\right)u(t)$

2. $\dfrac{Cs}{RCs + 1}$

3. (c)(i) $2e^{-t}(\sin 2t)u(t)$ (ii) $\left(-\frac{4}{5}e^{-t}\cos 2t - \frac{2}{5}e^{-t}\sin 2t + \frac{4}{5}\right)u(t)$